

MS4215: Advanced Data Analysis

Week 2B: Multiple Linear Regression

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Learning outcomes

- Explain why fitting separate simple regressions can mislead and how MLR corrects this
- Interpret partial regression coefficients as the effect of one predictor holding all others constant
- Read the MLR ANOVA table and interpret the global F-test and individual t-tests
- Distinguish R^2 from Adjusted R^2 and explain when Adjusted R^2 is preferred
- Fit and interpret a multiple linear regression model in R using `lm()`

From one predictor to many

Where we left off

Week 02A recap

We can now:

- Fit a simple regression: `lm(y ~ x)`
- Interpret the intercept and slope
- Decompose variance:
 $SST = SSR + SSE$
- Check LINE assumptions with
`plot(lm_object)`

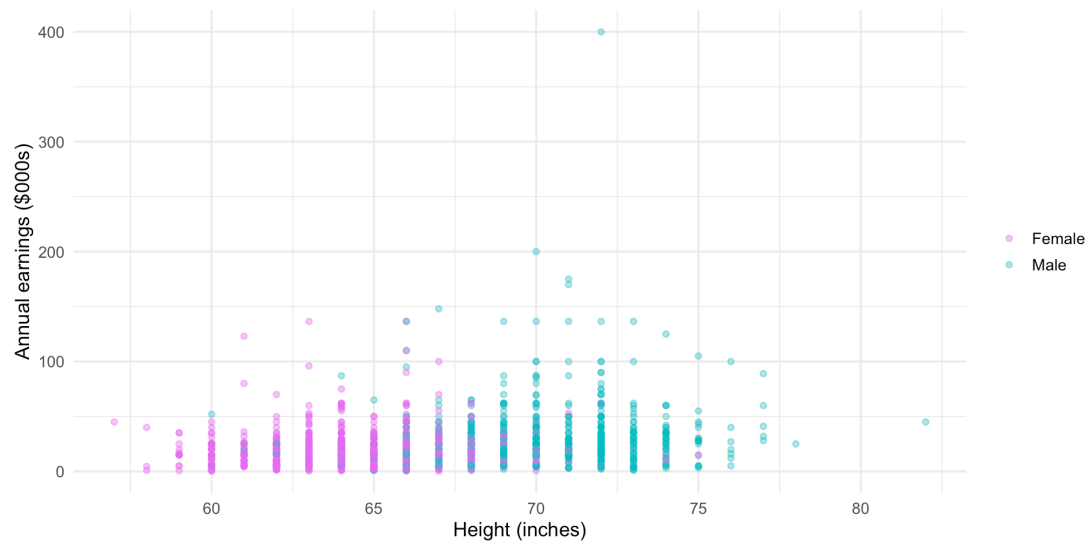
The question for today

What if a single predictor isn't the whole story?

What if ignoring other variables actually **distorts** the coefficient we estimate?

Today: extending SLR to **k predictors**, and why it matters even when each simple regression looks fine.

A question about earnings



earnings.csv : 1,816 US adults

Variable	Description
earn	Annual earnings (\$)
height	Height (inches)
gender	1 = male, 0 = female
education	Years of education
age	Age (years)

Question: Do taller people earn more? And does that depend on gender?

Why one regression at a time misleads

```
1 m_height      <- lm(earn ~ height, data = earnings)
2
3 m_height_gender <- lm(earn ~ height + gender_male, data = earnings)
```

SLR: `earn ~ height`

Height is significant. Taller people appear to earn more.

MLR: `earn ~ height + gender_male`

The height coefficient **changes** once gender is controlled.

Warning

Men tend to be taller *and* to earn more. A regression on height alone picks up part of the gender effect.

Holding gender constant, the independent height effect looks different.

This is what MLR does = it gives each predictor's contribution *separately*.

We will return to this in **Week 5A** (omitted variable bias) and **Week 6A** (moderation).

The problem with separate SLRs

If we have three potential predictors, what happens if we run three separate simple regressions?

Warning

Two problems:

1. With three separate prediction equations, there is no single prediction for a new observation when all three predictor values are given.
2. Each of the three regression equations ignores the other two predictors in forming estimates for the regression coefficients. If the predictors are correlated with each other, this can lead to **very misleading estimates**.

The solution: fit them all in one model = **Multiple Linear Regression**.

The advertising dataset

1 company. 200 markets.

For each: total product sales and three advertising budgets.

Variable	Role	Description
TV	predictor	TV advertising spend (\$000s)
radio	predictor	Radio advertising spend (\$000s)
newspaper	predictor	Newspaper advertising spend (\$000s)
sales	response	Product sales (thousands of units)

Research question: Which media drive sales, and by how much?

Running three separate SLRs

A natural first attempt: fit each predictor on its own:

Predictor	Coefficient	Units sold per \$1,000 spend	p-value	Verdict
TV	0.0475	~48 units	< 0.0001	Significant ✓
radio	0.203	~203 units	< 0.0001	Significant ✓
newspaper	0.055	~55 units	0.001	Significant ✓

All three look useful. Should we conclude newspaper advertising drives sales?

The correlation between predictors

Correlation matrix for the advertising data:

	TV	radio	newspaper	sales
TV	1.000	0.055	0.057	0.782
radio	—	1.000	0.354	0.576
newspaper	—	—	1.000	0.228
sales	—	—	—	1.000

Note

Radio and newspaper are moderately correlated ($r = 0.35$).

In markets where radio spend is high, newspaper spend also tends to be high.

When newspaper appears to predict sales in the SLR, it may be picking up radio's effect.

The MLR answer

Fit all three predictors in a single model:

	Coefficient	Std. Error	t	p-value
Intercept	2.939	0.312	9.42	< 0.0001
TV	0.046	0.001	32.81	< 0.0001
radio	0.189	0.009	21.89	< 0.0001
newspaper	-0.001	0.006	-0.18	0.86

Warning

Newspaper: **significant in SLR** ($p = 0.001$), **not significant in MLR** ($p = 0.86$).

Controlling for radio, newspaper adds nothing.

MLR separated what SLR conflated.

The MLR population model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

- k predictors, n observations, $k + 1$ unknown parameters
- We estimate $\beta_0, \beta_1, \dots, \beta_k$ with $b_0, b_1, \dots, b_k$ using ordinary least squares (OLS)

For the Advertising example ($k = 3$):

$$\text{sales} = \beta_0 + \beta_1 \text{TV} + \beta_2 \text{radio} + \beta_3 \text{newspaper} + \varepsilon$$

The fitted model gives: $\widehat{\text{sales}} = 2.939 + 0.046 \text{TV} + 0.189 \text{radio} - 0.001 \text{newspaper}$

Interpreting partial slopes

b_i = the estimated change in y for a **one-unit increase in x_i** , **holding all other predictors constant**

The coefficient for TV represents the average increase in product sales associated with increasing TV spending by \$1,000 **while holding radio and newspaper fixed**.

From the Advertising MLR:

- **TV:** every extra \$1,000 $\rightarrow$ +46 units sold (radio, newspaper held constant)
- **radio:** every extra \$1,000 $\rightarrow$ +189 units sold (TV, newspaper held constant)
- **newspaper:** no independent effect once TV and radio are controlled

Interpretation

A company models monthly sales:

$$\widehat{\text{Sales}} = 100 + 5.4 \times \text{Advertising} - 2.5 \times \text{Price}$$

Advertising coefficient: +5.4

For each additional unit of advertising spend, **holding price constant**, expected sales increase by 5.4.

The price stays fixed, only advertising varies.

Price coefficient: -2.5

For each additional unit of price, **holding advertising constant**, expected sales decrease by 2.5.

A negative sign here makes economic sense: higher price → lower demand.

Note

The sign of each coefficient is only meaningful when the other predictors are held fixed.

This is why we cannot interpret MLR coefficients one at a time as if from separate SLRs.

MLR assumptions

The same **LINE** conditions apply to the errors ε :

	Assumption	
L	Linearity	$E(\varepsilon_i) = 0$; the relationship is linear in each predictor
I	Independence	Errors are independent across observations
N	Normality	$\varepsilon_i \sim N(0, \sigma^2)$
E	Equal variance	$\text{Var}(\varepsilon_i) = \sigma^2$ for all i

Plus one new condition:

No perfect multicollinearity: no predictor can be an exact linear combination of the others. This ensures $(X'X)^{-1}$ exists.

Imperfect multicollinearity (e.g. radio-newspaper $r = 0.35$) is common and manageable. It inflates standard errors but does not break OLS. We will return to this shortly.

Estimating the MLR Model

Why matrix notation?

With k predictors and n observations, writing every equation out is unwieldy:

$$Y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \cdots + \beta_k x_{1k} + \varepsilon_1$$

$$Y_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \cdots + \beta_k x_{2k} + \varepsilon_2$$

$$\vdots$$

$$Y_n = \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \cdots + \beta_k x_{nk} + \varepsilon_n$$

One compact expression covers all n equations at once:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Note

This is exactly what R computes internally when you call `lm()`.

The design matrix

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Dimensions:

- $\mathbf{Y}$: $n \times 1$ (response vector)
- $\mathbf{X}$: $n \times (k + 1)$ (design matrix)
- $\boldsymbol{\beta}$: $(k + 1) \times 1$ (parameter vector)
- $\boldsymbol{\varepsilon}$: $n \times 1$ (error vector)

The design matrix:

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{pmatrix}$$

First column: all 1s (for the intercept β_0)

In R: `X <- cbind(rep(1, n), x1, x2)`

The matrix model, expanded

For $n = 3$ observations, $k = 2$ predictors:

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ 1 & x_{31} & x_{32} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{pmatrix}$$

Each **row** unpacks to one observation's equation:

$$y_i = 1 \beta_0 + x_{i1} \beta_1 + x_{i2} \beta_2 + \varepsilon_i$$

The compact form $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ carries all n equations simultaneously.

The OLS objective

We want to choose **b** to **minimise the sum of squared residuals**:

$$SSE = \hat{\varepsilon}' \hat{\varepsilon} = (\mathbf{Y} - \mathbf{Xb})'(\mathbf{Y} - \mathbf{Xb})$$

Differentiating with respect to **b** and setting to zero:

$$\frac{\partial SSE}{\partial \mathbf{b}} = -2\mathbf{X}'(\mathbf{Y} - \mathbf{Xb}) = \mathbf{0}$$

This yields the **normal equations**:

$$\mathbf{X}'\mathbf{X} \mathbf{b} = \mathbf{X}'\mathbf{Y}$$

The OLS solution

If $\mathbf{X}'\mathbf{X}$ is invertible, solving the normal equations gives:

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{Y}$$

This **generalises the SLR formula**, for $k = 1$ it reduces to:

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Verify in R:

```
1 X <- cbind(rep(1, nrow(advertising)), advertising$TV, advertising$radio, advertising$newspaper)
2 y <- advertising$sales
3 b_manual <- solve(t(X) %*% X) %*% t(X) %*% y
4 b_manual
5 coef(lm(sales ~ TV + radio + newspaper, data = advertising)) #identical
```

The hat matrix

Substituting $\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ into the fitted values:

$$\hat{\mathbf{Y}} = \mathbf{X}\mathbf{b} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \mathbf{H}\mathbf{Y}$$

where $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$

$\mathbf{H}$ is called the **hat matrix**. It “puts the hat on” $\mathbf{Y}$ to produce $\hat{\mathbf{Y}}$.

The diagonal entries h_{ii} measure how much each observation influences its own fitted value.

We will return to these in **Week 5B** when we cover leverage and influential observations.

b is unbiased

The OLS estimator $\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ is an **unbiased** estimator of β .

The key step: since $E(\varepsilon) = \mathbf{0}$ by assumption,

$$E(\mathbf{Y}) = \mathbf{X}\beta$$

Substituting this in gives:

$$E(\mathbf{b}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta = \beta \quad \checkmark$$

On average across many samples, **b** is exactly right.

Variance of $\mathbf{b}$

The sampling variability of $\mathbf{b}$ is captured by:

$$\text{Cov}(\mathbf{b}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

- The diagonal element $[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$ gives $\text{Var}(b_j)$
- $\text{SE}(b_j) = s \sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$ where $s = \sqrt{\text{MSE}}$

These are exactly the **Std. Error** column values in `summary(lm_object)`.

Note

When $(\mathbf{X}'\mathbf{X})^{-1}$ has large diagonal elements (caused by near-multicollinearity) the standard errors inflate.

This is how multicollinearity makes individual t-tests unreliable.

Estimating σ^2

We estimate σ^2 (the error variance) with:

$$s^2 = \text{MSE} = \frac{\text{SSE}}{n - k - 1}$$

Degrees of freedom for error: $n - k - 1$

- n observations
- $k + 1$ parameters estimated ($\beta_0, \beta_1, \dots, \beta_k$)
- Leaves $n - k - 1$ degrees of freedom

This appears as “**Residual standard error**” ($s = \sqrt{\text{MSE}}$) in `summary()`.

Model	k	df for error	MSE
SLR	1	$n - 2$	$\text{SSE}/(n - 2)$
MLR	k	$n - k - 1$	$\text{SSE}/(n - k - 1)$

Separating random and systematic structure

You can decompose every linear model into two components:

Component	Expression	What it captures
Random structure	$\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$	How Y varies around its mean
Systematic structure	$\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$	How the mean depends on predictors

The expression $\mathbf{X}\boldsymbol{\beta}$ is called the **linear predictor**. It captures systematic sources of variation; the error term accounts for the rest.

The fitted values $\hat{\mathbf{Y}} = \mathbf{X}\mathbf{b}$ are our estimate of $\boldsymbol{\mu}$ = the systematic part of the model.

Note

This decomposition generalises directly to GLMs in Week 9. The only things that change are the distribution of Y and how $\boldsymbol{\mu}$ relates to $\mathbf{X}\boldsymbol{\beta}$.

Null and saturated models

Two extremes bracket every model we can fit:

Model	Parameters	Fitted values	What it says
Null	1	$\hat{\mu}_i = \bar{y}$ for all i	No predictor explains anything
Fitted MLR	$k + 1$	$\hat{\mu}_i = \mathbf{x}_i' \mathbf{b}$	Systematic structure captured
Saturated	n	$\hat{\mu}_i = y_i$ for each i	Perfect fit. One parameter per observation

“Obviously the null and saturated models are not very useful by themselves. Most statistical models of interest lie somewhere in between.”

Note

In Week 9, these become the benchmarks for **deviance**: how far does the fitted model fall short of the saturated model? How far does the null model fall short? The distance between them measures how much our predictors explain.

Evaluating the MLR Model

Variance decomposition, revisited

$$SST = SSR + SSE$$

This still holds for MLR. Now **SSR is the variation explained by all k predictors together.**

Degrees of freedom split:

Source	df
Regression (all k predictors)	k
Error	$n - k - 1$
Total	$n - 1$

With $k = 1$ (SLR): df for regression = 1 and df for error = $n - 2$, exactly as in Week 02A.

Each predictor added to the model “uses up” one degree of freedom from error.

The MLR ANOVA table

Source	df	SS	MS	F
Regression	k	SSR	$MSR = SSR/k$	MSR/MSE
Error	$n - k - 1$	SSE	$MSE = SSE/(n - k - 1)$	
Total	$n - 1$	SST		

In R:

```
1 summary(ireland_mlr) # F-statistic at the bottom = overall F test
```

Call:

```
lm(formula = tmax_c ~ tmin_c + prcp_mm, data = ireland)
```

Residuals:

```
    Min       1Q   Median       3Q      Max
-7.049 -1.966 -0.325  1.700 13.890
```

Coefficients:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.828142   0.037231  210.26  <2e-16 ***
tmin_c       0.805854   0.004210  191.40  <2e-16 ***
prcp_mm     -0.072616   0.003438  -21.12  <2e-16 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 2.705 on 22038 degrees of freedom
(20149 observations deleted due to missingness)
```

```
Multiple R-squared:  0.6252    Adjusted R-squared:  0.6251
```

The F-test: Testing the overall model

$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$ (no predictors are useful)

$H_1 : \text{at least one } \beta_i \neq 0$

Under H_0 : $F = \frac{\text{MSR}}{\text{MSE}} \sim F(k, n - k - 1)$

Reject H_0 at level α if $F > F_\alpha(k, n - k - 1)$.

Advertising MLR: $F = 570, R^2 = 0.897$

Compare: SLR of sales on TV alone had $F = 312, R^2 = 0.612$.

Adding radio and newspaper (controlling properly) explains far more variance and produces a much stronger overall test.

ANOVA example

Given: $n = 154$, $k = 3$

Source	df	SS	MS	F
Regression	—	6025.5	—	2.53
Error	—	—	—	
Total	—	—		

Can you fill in the rest of the table?

Given: $n = 154$, $k = 3$, $SSR = 6025.5$, $F = 2.53$

Using only these four values, fill in every blank cell:

1. df for Regression, Error, Total
2. $MSR = SSR \div df_{\text{reg}}$
3. $MSE = MSR \div F$
4. $SSE = MSE \times df_{\text{error}}$
5. $SST = SSR + SSE$

ANOVA example: Solution

Source	df	SS	MS	F
Regression	3	6025.5	2008.5	2.53
Error	150	119,085	793.9	
Total	153	125,111		

Working:

- $df_{\text{reg}} = k = 3$; $df_{\text{error}} = n - k - 1 = 150$; $df_{\text{total}} = n - 1 = 153$
- $MSR = 6025.5/3 = 2008.5$
- $MSE = 2008.5/2.53 = 793.9$
- $SSE = 793.9 \times 150 = 119,085$
- $SST = 6025.5 + 119,085 = 125,111$

Full model vs reduced model: A unified view

The overall F-test we just saw is a special case of a general framework. Every F-test in regression compares a **full model** and a **reduced model**:

$$F = \frac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}}) / q}{\text{SSE}_{\text{full}} / (n - p)}$$

where q = number of parameters dropped, p = number of parameters in the full model.

Test	Reduced model	Full model	q
Overall F	intercept only (β_0)	all k predictors	k
Partial F	subset of predictors	larger set of predictors	difference in k

In R: `anova(reduced_model, full_model)` performs this comparison directly.

Note

One framework covers both the overall test and tests of subsets of predictors. The only difference is which parameters are constrained to zero.

R^2 for MLR

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

Same formula as SLR. Now measures the proportion of variance in Y explained by **all k predictors together**.

Model	R^2	Interpretation
SLR: sales ~ TV	0.612	TV alone explains 61.2% of sales variance
MLR: sales ~ TV + radio + newspaper	0.897	All three explain 89.7% of sales variance

R^2 increased from 0.612 to 0.897 by adding radio and newspaper = a substantial improvement.

The problem with R^2

Warning

R^2 never decreases when you add a predictor! Even a useless one!

Why? Adding any new predictor gives OLS another “lever” to reduce SSE. Even if the predictor is pure random noise, SSE can only decrease or stay the same. So $R^2 = 1 - \text{SSE}/\text{SST}$ can only increase or stay flat.

Consequence: if we compare a 3-predictor model with a 10-predictor model using R^2 , the 10-predictor model always wins. Even if the extra 7 predictors are useless.

We need a measure that penalises for complexity.

Adjusted R^2

$$R_{\text{adj}}^2 = 1 - \frac{\text{SSE}/(n - k - 1)}{\text{SST}/(n - 1)} = 1 - (1 - R^2) \frac{n - 1}{n - k - 1}$$

The penalty term $\frac{n - 1}{n - k - 1} > 1$ grows as k increases.

- Adding a **useful** predictor: SSE drops enough to outweigh the df penalty $\rightarrow R_{\text{adj}}^2$ increases
- Adding a **useless** predictor: SSE barely changes, df penalty kicks in $\rightarrow R_{\text{adj}}^2$ **decreases**

Note

R_{adj}^2 is always $\leq R^2$. For large n relative to k , the two are close.

For small n with many predictors, the penalty is substantial.

R^2 vs Adjusted R^2 : When to use each

	R^2	Adjusted R^2
Use for	Describing fit of a given model	Comparing models with different numbers of predictors
Changes when adding a predictor	Always increases (or stays flat)	Can increase or decrease
Penalises complexity	No	Yes
Reported by <code>summary()</code>	✓	✓

When using adjusted R^2 as the decision criterion, we seek to eliminate or add predictors depending on whether they lead to the **largest improvement in adjusted R^2** , and we stop when adding or eliminating another predictor does not lead to further improvement.

Tests for individual predictors

For each b_j in the model:

$H_0 : \beta_j = 0$ (given all other predictors in the model)

$$t = \frac{b_j}{\text{SE}(b_j)}, \quad t \sim t(n - k - 1) \text{ under } H_0$$

From `summary()` in R:

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.939	0.312	9.42	<2e-16 ***
TV	0.046	0.001	32.81	<2e-16 ***
radio	0.189	0.009	21.89	<2e-16 ***
newspaper	-0.001	0.006	-0.18	0.857

Note

The p-value for each predictor is **conditional on all others being in the model**. This is why newspaper's p-value changed so dramatically from SLR to MLR.

F-test vs t-tests: What each tells you

Overall F-test

H_0 : none of the predictors are useful ($\beta_1 = \dots = \beta_k = 0$)

Tests the model as a whole.

Advertising: $F = 570, p < 0.0001$

The model works.

Individual t-tests

H_0 : this predictor adds nothing given the others ($\beta_j = 0$)

Tests each predictor in context.

Advertising: newspaper $t = -0.18,$
 $p = 0.86$

This predictor is redundant.

Note

A significant overall F + some non-significant t tests is a signal: the model is useful, but some predictors may be redundant and could be removed.

The overall F can be significant even when individual predictors are not.

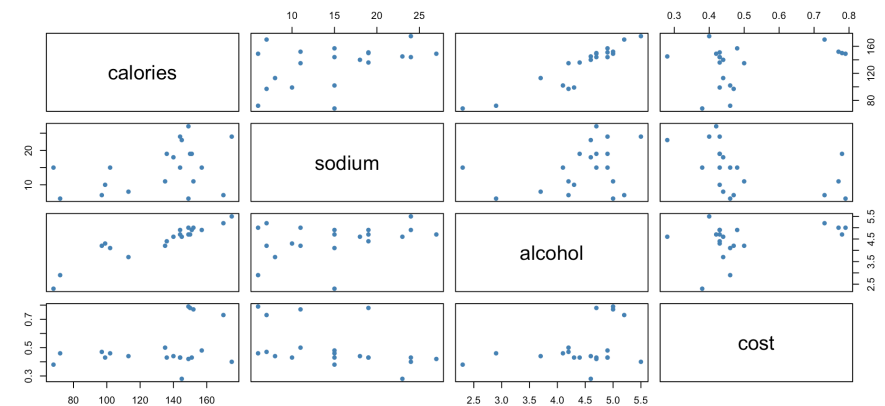
MLR in Practice

Beer data: Loading and exploring

```
1 beer <- read_csv("data/beer.csv")
2 head(beer, 20)
```

```
# A tibble: 20 × 5
  beer          calories sodium alcohol  cost
  <chr>         <dbl>   <dbl>   <dbl> <dbl>
1 Budweiser      144     15     4.7  0.43
2 Schlitz        151     19     4.9  0.43
3 Lowenbrau       157     15     4.9  0.48
4 Kronenbourg    170      7     5.2  0.73
5 Heineken       152     11      5   0.77
6 Old_Milwaukee  145     23     4.6  0.28
7 Augsberger     175     24     5.5  0.4
8 Strohs_Bohemian_Style 149     27     4.7  0.42
9 Miller_Lite     99     10     4.3  0.43
10 Budweiser_Light 113      8     3.7  0.44
11 Coors          140     18     4.6  0.44
12 Coors_Light    102     15     4.1  0.46
13 Michelob_Light 135     11     4.2  0.5
14 Becks          150     19     4.7  0.78
15 Kirin          149      6      5   0.79
16 Pabst_Extra_Light 68     15     2.3  0.38
17 "
```

```
1 pairs(beer[, 2:5],
2       pch = 16,
3       col = "steelblue")
```



20 brands of beer. **Which predictors look most linearly related to alcohol content?**

lm() for MLR

The `lm()` formula interface extends naturally to multiple predictors:

```
1 # Add predictors with +  
2 lm(alc0hol ~ sodium + calories + cost, data = beer)
```

Call:

```
lm(formula = alc0hol ~ sodium + calories + cost, data = beer)
```

Coefficients:

(Intercept)	sodium	calories	cost
1.433610	-0.009507	0.024120	-0.090475

Key functions for the fitted model object:

Function	Output
<code>summary()</code>	Coefficients, SEs, t-tests, R^2 , Adj R^2 , F-statistic
<code>anova()</code>	Sequential SS for each predictor
<code>anova(fit1, fit2)</code>	Partial F-test comparing two nested models
<code>confint()</code>	Confidence intervals for each β_j
<code>plot()</code>	Four-panel LINE diagnostics

Beer: SLR baseline

```
1 beer_slr <- lm(alcohol ~ calories, data = beer)
2 summary(beer_slr)
```

Call:

```
lm(formula = alcohol ~ calories, data = beer)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.65066	-0.13434	-0.06234	0.12407	0.63242

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.378058	0.312769	4.406	0.000341	***
calories	0.023126	0.002306	10.030	8.53e-09	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3041 on 18 degrees of freedom

Multiple R-squared: 0.8482, Adjusted R-squared: 0.8398

F-statistic: 100.6 on 1 and 18 DF, p-value: 8.532e-09

Fitted equation: $\widehat{\text{alcohol}} = 1.378 + 0.0231 \text{ calories}$

$R^2 = 0.848$, $\text{RSE} = 0.304$, $n = 20$

Calories alone explains **84.8% of the variation in alcohol content**. A strong start, but can sodium and cost add anything?

Beer: Fitting the MLR

```
1 beer_mlr <- lm(alcohol ~ sodium + calories + cost, data = beer)
2 summary(beer_mlr)
```

Call:

```
lm(formula = alcohol ~ sodium + calories + cost, data = beer)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.59680	-0.14376	-0.02914	0.10315	0.61245

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.433610	0.407330	3.520	0.00284	**
sodium	-0.009507	0.016317	-0.583	0.56823	
calories	0.024120	0.003367	7.164	2.25e-06	***
cost	-0.090475	0.709423	-0.128	0.90011	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.318 on 16 degrees of freedom

Multiple R-squared: 0.8525, Adjusted R-squared: 0.8248

Note

Model: $\widehat{\text{alcohol}} = 1.434 - 0.0095 \text{ sodium} + 0.0241 \text{ calories} - 0.090 \text{ cost}$

$R^2 = 0.853$, Adjusted $R^2 = 0.825$, RSE = 0.318, $F = 30.82$ (df 3, 16), $n = 20$

Beer: Reading the partial slopes

$$\widehat{\text{alcohol}} = 1.434 - 0.0095 \text{ sodium} + 0.0241 \text{ calories} - 0.090 \text{ cost}$$

Predictor	Coeff	t	Interpretation
sodium	-0.0095	-0.58	Each extra mg of sodium → -0.0095% alcohol (calories, cost fixed). Not significant
calories	+0.0241	7.16	Each extra calorie → +0.024% alcohol (sodium, cost fixed). Strong effect
cost	-0.090	-0.13	Each extra dollar → -0.090% alcohol (sodium, calories fixed). Not significant

Note

Calories is the only predictor that matters. Sodium and cost add nothing once calories is in the model. The tight calorie–alcohol link leaves no independent signal for the other predictors to exploit.

Beer: `anova()` for sequential SS

```
1 anova(beer_mlr)

Analysis of Variance Table

Response: alcohol
      Df Sum Sq Mean Sq F value    Pr(>F)
sodium   1  1.1334   1.1334  11.2074  0.004087 **
calories  1  8.2150   8.2150  81.2360 1.142e-07 ***
cost      1  0.0016   0.0016   0.0163  0.900108
Residuals 16  1.6180   0.1011
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

`anova()` on a single model object gives **sequential (Type I) SS**. The contribution of each predictor given those entered before it:

- SS for sodium: sodium's contribution entering first
- SS for calories: calories' contribution *given sodium already in the model*
- SS for cost: cost's contribution *given sodium and calories already in the model*

Note

Sequential SS depends on the order predictors appear in the formula. The overall F-statistic in `summary()` does not.

Comparing nested models with `anova()`

To ask “do sodium and cost add anything over calories alone?”, use `anova()` with two models:

```
1 beer_reduced <- lm(alcohol ~ calories, data = beer)
2 beer_full    <- lm(alcohol ~ calories + sodium + cost, data = beer)
3 anova(beer_reduced, beer_full)
```

Analysis of Variance Table

Model 1: alcohol ~ calories

Model 2: alcohol ~ calories + sodium + cost

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	18	1.6647				
2	16	1.6180	2	0.046689	0.2308	0.7965

This is the **partial F-test**: $H_0: \beta_{\text{sodium}} = \beta_{\text{cost}} = 0$ given calories is in the model.

$F = 0.23$, $p = 0.80$: sodium and cost together add nothing beyond what calories already explains.

Note

Rule: Do the reduced model first, full model second.

The output shows the additional SS, df, F-statistic, and p-value for the extra predictors.

Beer: R^2 vs Adjusted R^2

Model	R^2	Adj R^2
SLR: <code>alcohol ~ calories</code>	0.848	0.840
MLR: <code>alcohol ~ sodium + calories + cost</code>	0.853	0.825

Interpretation:

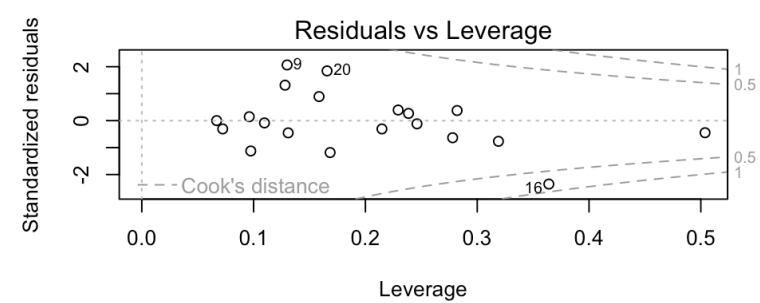
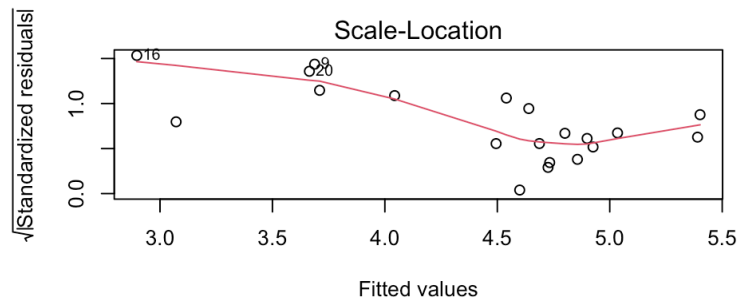
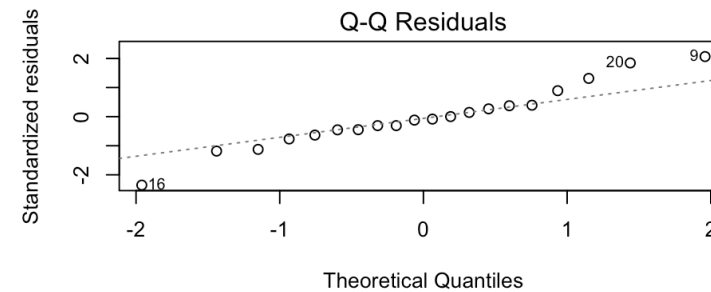
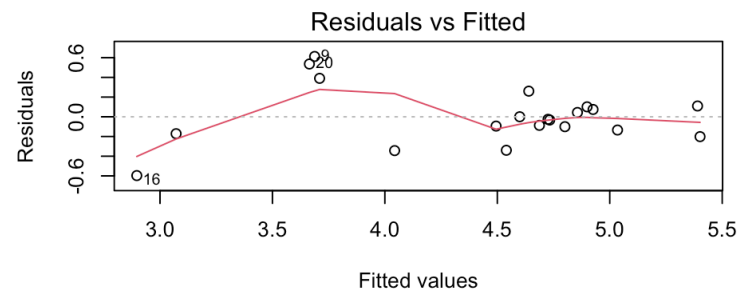
Adding sodium and cost barely moves R^2 ($0.848 \rightarrow 0.853$, +0.5%).

Adjusted R^2 **falls** ($0.840 \rightarrow 0.825$): the extra predictors do not earn their place. The df penalty outweighs the trivial improvement in fit.

This confirms what the t-tests already told us: sodium and cost add nothing.

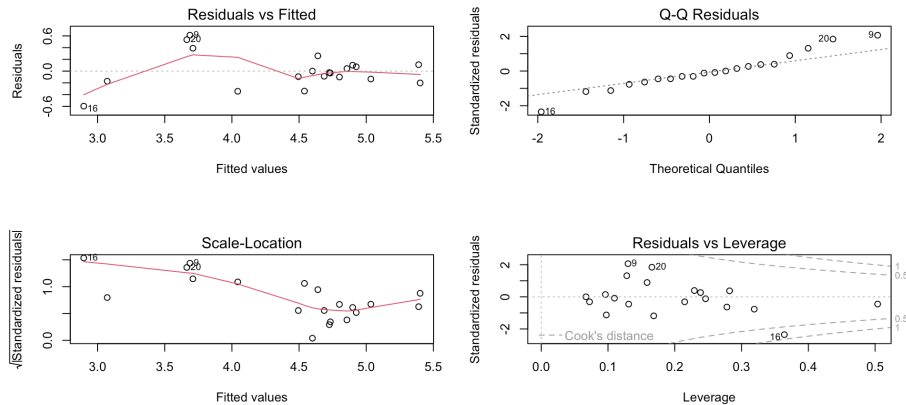
Beer: Diagnostics

```
1 par(mfrow = c(2, 2))
2 plot(beer_mlr)
```



Beer: Diagnostics

```
1 par(mfrow = c(2, 2))
2 plot(beer_mlr)
```



Things to look for:

- **Residuals vs Fitted:** no systematic curve (linearity ✓)
- **Normal Q-Q:** points close to the diagonal (normality ✓)
- **Scale-Location:** roughly flat red line (equal variance ✓)
- **Residuals vs Leverage:** any points with high Cook's distance?

Four-panel LINE check for the Beer MLR. With $n = 20$, patterns are approximate: small datasets produce noisy diagnostic plots.

Extending the Irish weather model

In Week 02A we fitted: `lm(tmax_c ~ tmin_c, data = ireland)`: $R^2 = 0.618$

Does precipitation add predictive value?

```
1 ireland_mlr <- lm(tmax_c ~ tmin_c + prcp_mm, data = ireland)
2 summary(ireland_mlr)
```

Call:

```
lm(formula = tmax_c ~ tmin_c + prcp_mm, data = ireland)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.049	-1.966	-0.325	1.700	13.890

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.828142	0.037231	210.26	<2e-16 ***
tmin_c	0.805854	0.004210	191.40	<2e-16 ***
prcp_mm	-0.072616	0.003438	-21.12	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.705 on 22038 degrees of freedom

(20149 observations deleted due to missingness)

Multiple R-squared: 0.6252, Adjusted R-squared: 0.6251

Extending the Irish weather model

In Week 02A we fitted: `lm(tmax_c ~ tmin_c, data = ireland)`: $R^2 = 0.618$

Does precipitation add predictive value?

Fitted equation: $\widehat{tmax} = 7.828 + 0.806 \times tmin - 0.073 \times prcp_mm$
 $R^2 = 0.625$, $Adj\ R^2 = 0.625$, $n = 22,041$

Note

`prcp\_mm` is statistically significant (with $n = 22,041$, almost any effect will be), but adds only +0.7 percentage points to R^2 .
Statistical significance $\neq$ practical importance.

Variable selection: The problem

With many potential predictors, which do we include?

Too many predictors:

- Includes irrelevant variables (the newspaper problem)
- Inflates standard errors
- Harder to interpret
- Risk of overfitting

Too few predictors:

- Omits relevant variables → biased coefficients
- Leaves systematic patterns in residuals
- Lower predictive power

Note

The goal: find the model that balances fit and parsimony.

Adjusted R^2 is one criterion for this tradeoff.

R functions

Function	Purpose
<code>lm(y ~ x1 + x2 + x3, data = df)</code>	Fit multiple linear regression
<code>summary(model)</code>	Coefficients, Adj R ² , F-test
<code>anova(model)</code>	Sequential ANOVA table
<code>anova(m1, m2)</code>	Partial F-test for nested models
<code>confint(model)</code>	Confidence intervals for coefficients
<code>predict(model, newdata)</code>	Fitted values or predictions
<code>plot(model)</code>	Four LINE diagnostic plots

Summary

- Separate SLRs mislead when predictors correlate. MLR controls for all predictors simultaneously
- OLS solution: $\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$
- Partial slopes: “holding all other variables constant”
- F-test: overall model ($H_0: \text{all } \beta_j = 0$); t-tests: individual predictors
- R^2 always increases; **Adjusted R^2** penalises for extra predictors
- `anova(fit1, fit2)` : partial F-test for nested models